Digital Package Weight Verification System Using System Verilog (RTL --> Synth)

Team_EDITH:

Karthiswaran R

Sathiya Jeeva M

TamilRaja S B

Electronics Engineering (VLSI Design and Technology)

IIIrd Year

K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(Autonomous)

Problem Statement

- In modern manufacturing and packaging industries, ensuring that every packaged product meets the required weight standards is critical for quality control, regulatory compliance, and customer satisfaction.
- However, many packaging systems still face issues such as:
- Incorrect product weight
- Manual inspection errors
- Production inefficiencies
- Product wastage
- Customer complaints and regulatory penalties
- There is a need for an automated digital system that can verify package weights in real time and reject faulty packages.

Objectives

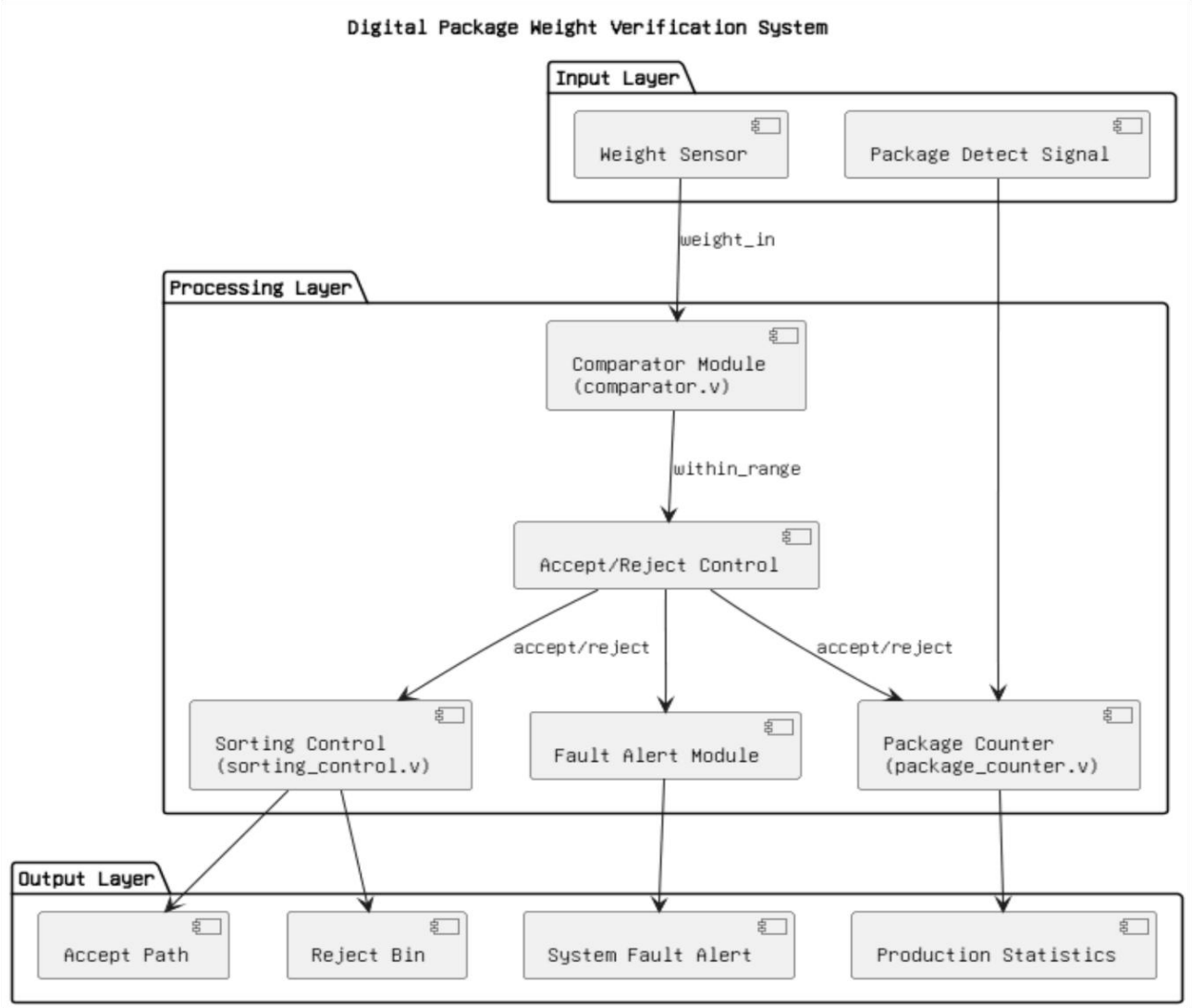
The main objectives of this project are:

- To design a digital controller for automated weight verification.
- To ensure packages fall within the acceptable weight range.
- To automatically accept or reject packages based on weight.
- To maintain real-time package counting and monitoring.
- To generate fault alerts for abnormal conditions.
- To enable efficient sorting in automated production lines.

System Architecture

- Key functional modules of the system include:
- Weight Comparator
  - Compares measured weight with allowed limits.
- Acceptance / Rejection Control
  - Determines whether the package is valid.
- Package Counter
  - Tracks total processed and rejected packages.
- Fault Alert System
  - Detects abnormal sensor readings.
- Sorting Control
  - Directs packages to the correct conveyor path.
- This modular architecture ensures scalability and reliability.

Block Diagram



Proposed Solution

We propose a Digital Package Weight Verification System that integrates:

- Weight comparison logic
- Acceptance and rejection control
- Automatic sorting signals
- Package counters
- Fault detection alerts
- The system continuously checks the weight of packages and ensures that only valid packages move forward in the production line.
- Invalid packages are automatically diverted to a rejection path.

Sustainability Assessment

Environmental Sustainability

- Reduces product wastage by ensuring accurate weight verification and minimizing rejected materials.
- Optimizes resource usage by preventing over-packaging and improving manufacturing efficiency.

Social Sustainability

- Ensures consistent product quality and builds consumer trust in packaged goods.
- Reduces manual inspection workload, improving worker safety and operational efficiency.

Economic Sustainability

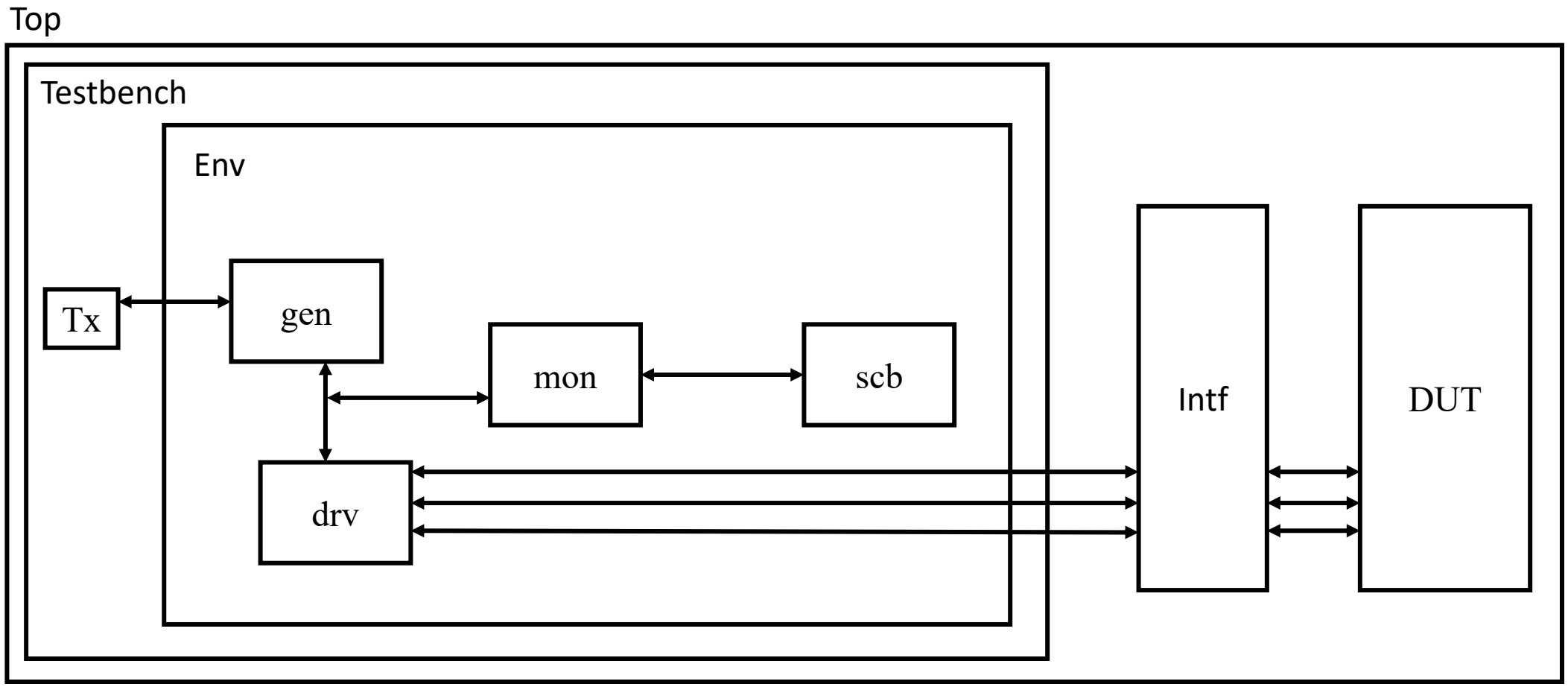
- Minimizes financial losses caused by incorrect product weights and product recalls.
- Provides a scalable and cost-effective solution that improves productivity in automated manufacturing lines.

Scalability & Future Scope

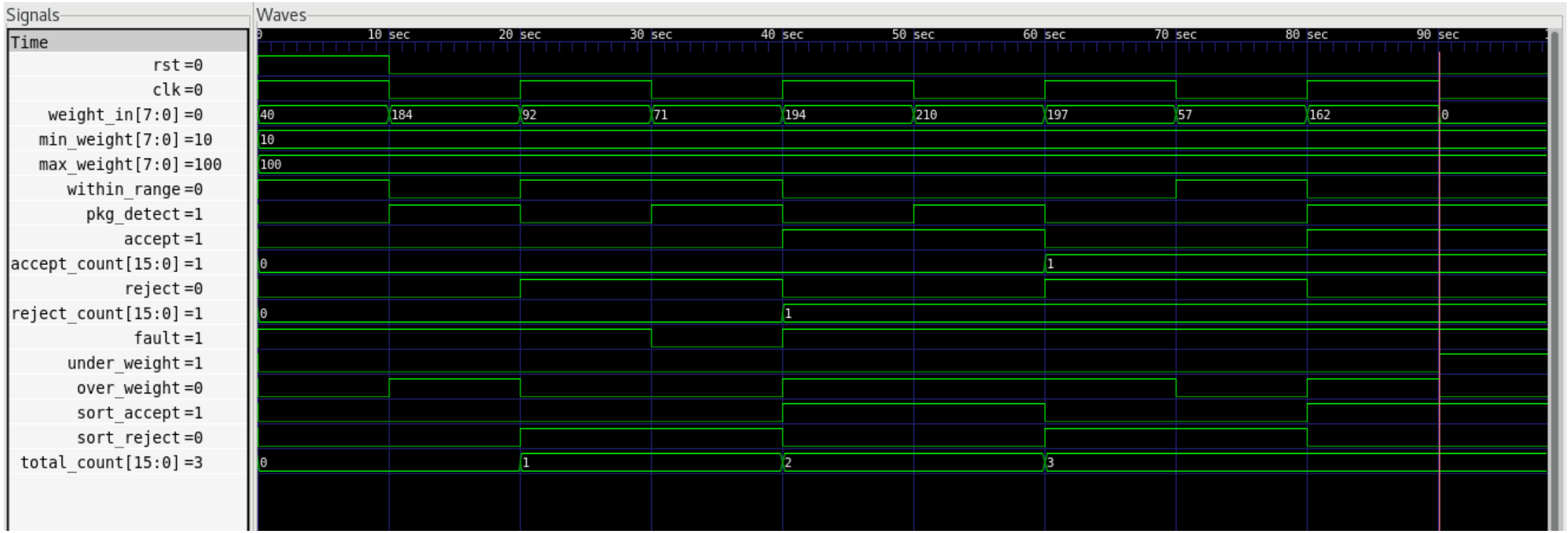
The system can be extended with:

- IoT-based production monitoring
- AI-based anomaly detection

SV Architecture



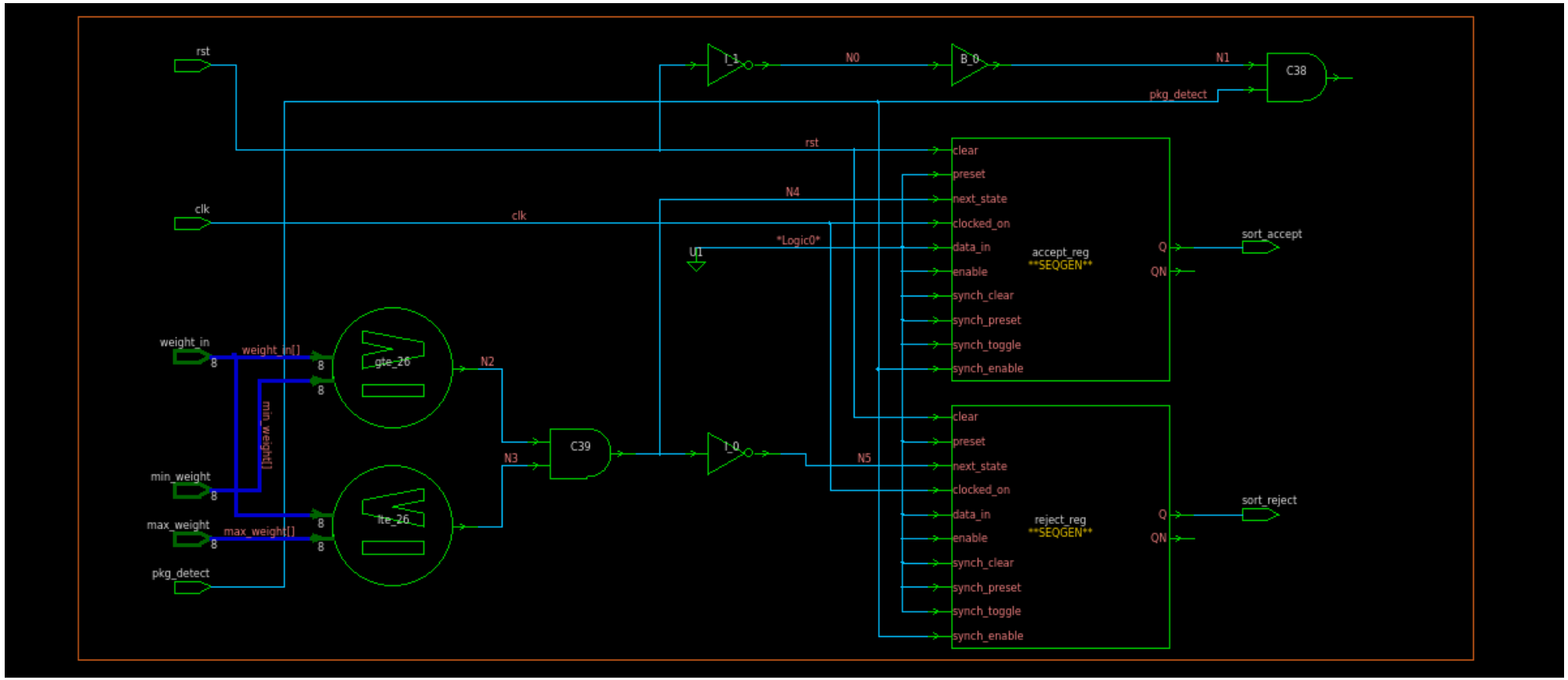
Functional Verification



Functional Verification Results

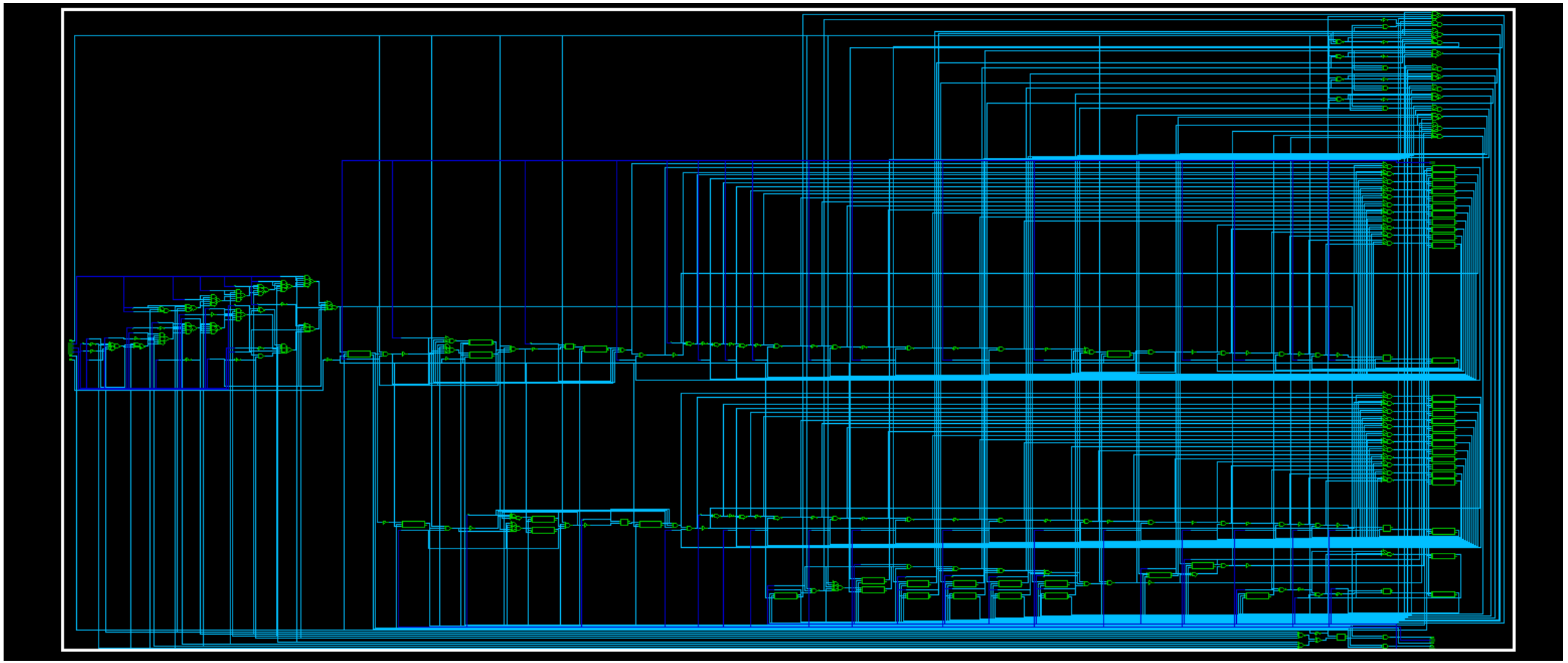
- RTL verified using System Verilog layered testbench architecture.
- Components used: Generator, Driver, Monitor, Scoreboard, Environment
- Randomized stimulus used to test different weight scenarios
- Verified cases:
  - Underweight packages
  - Valid weight packages
  - Overweight packages
  - Sensor fault conditions
- Scoreboard comparison shows zero mismatches

GTECH Netlist



Synthesis Block Diagram

Synthesized Netlist

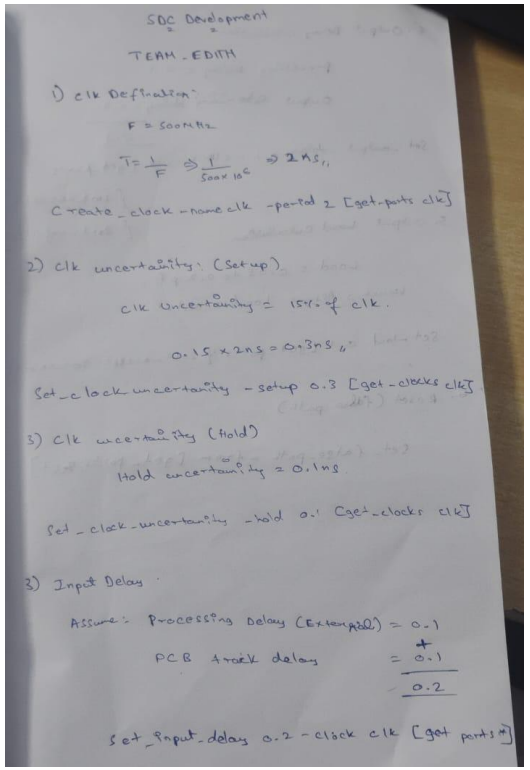


Specifications

Metric	Result
Target Frequency	500 MHz
Critical Path Delay	1.42 ns
Total Cells	~230
Design Area	~900
Setup Violations	0
Hold Violations	0
Verification Status	Pass

SDC Development for Optimization

- The previous SDC was over constrained, so we need to reframe a clock frequency and in addition we have added hold uncertainty to fix the 98 violating paths.



SDC Development

TEAM - EDITH

1) clk Defination:

F = 500MHz

$T = \frac{1}{F} \Rightarrow \frac{1}{500 \times 10^6} \Rightarrow 2ns$

Create_clock -name clk -period 2 [get_ports clk]

2) clk uncertainity: (Setup)

clk Uncertainity = 15% of clk.

0.15 × 2ns = 0.3ns

Set_clock_uncertainity -setup 0.3 [get_clocks clk]

3) Clk uncertainity (Hold)

Hold uncertainity = 0.1ns

Set_clock_uncertainity -hold 0.1 [get_clocks clk]

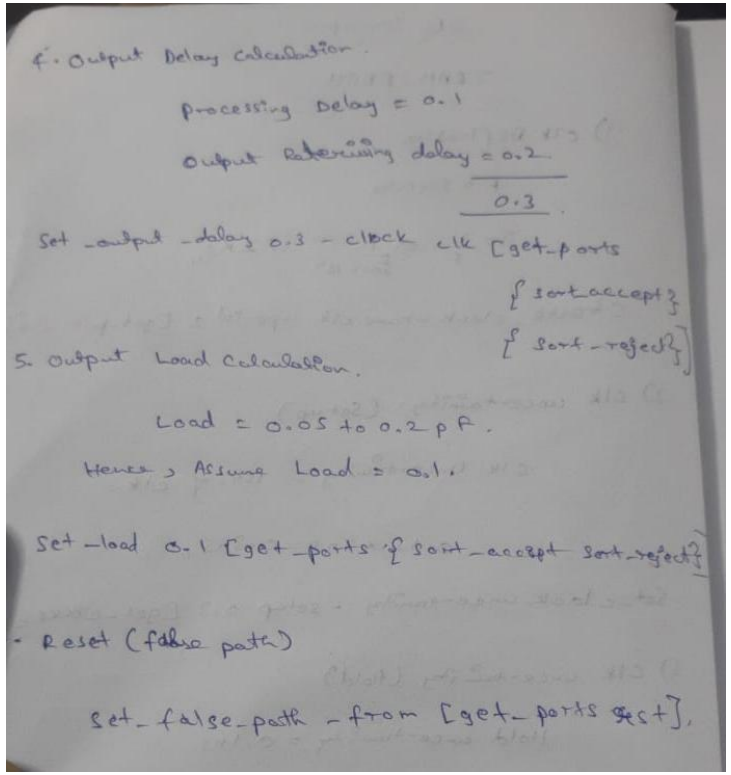
3) Input Delay

Assume: Processing Delay (Extrapolt) = 0.1

PCB track delay = 0.1

0.2

set_input_delay 0.2 -clock clk [get ports *]



4. Output Delay Calculation.

Processing Delay = 0.1

Output Referencing delay = 0.2

0.3

Set_output_delay 0.3 -clock clk [get_ports {sort_accept} {sort_reject}]

5. Output Load Calculation.

Load = 0.05 to 0.2 pF.

Hence, Assume Load = 0.1.

Set_load 0.1 [get_ports {sort_accept sort_reject}]

- Reset (false path)

set_false_path -from [get_ports rst].

Timing Report Analysis

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : weight_verification_top
Version: W-2024.09-SP5-3
Date   : Sat Mar 14 06:29:53 2026
****************************************

Operating Conditions: nom_pvt   Library: dw_foundation.sldb
Wire Load Model Mode: top

  Startpoint: ctrl/accept_reg
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: sort_accept
            (output port clocked by clk)
  Path Group: clk
  Path Type: max

  Point                                                    Incr       Path
  -----------------------------------------------------------------------
  clock clk (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  ctrl/accept_reg/clocked_on (**SEQGEN**)                  0.00       0.00 r
  ctrl/accept_reg/Q (**SEQGEN**)                           0.00       0.00 r
  ctrl/accept (accept_reject_control)                      0.00       0.00 r
  sort_ctrl/accept (sorting_control)                       0.00       0.00 r
  sort_ctrl/B_0/Z (GTECH_BUF)                              0.00       0.00 r
  sort_ctrl/C19/Z_0 (*SELECT_OP_3.1_3.1_1)                 0.00       0.00 r
  sort_ctrl/sort_accept (sorting_control)                  0.00       0.00 r
  sort_accept (out)                                        0.00       0.00 r
  data arrival time                                                   0.00

  clock clk (rise edge)                                   10.00      10.00
  clock network delay (ideal)                              0.00      10.00
  clock uncertainty                                       -0.30       9.70
  output external delay                                   -0.30       9.40
  data required time                                                  9.40
  -----------------------------------------------------------------------
  data required time                                                  9.40
  data arrival time                                                   0.00
  -----------------------------------------------------------------------
  slack (MET)                                                         9.40
```

Before optimization

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : weight_verification_top
Version: W-2024.09-SP5-3
Date   : Sat Mar 14 06:26:08 2026
****************************************

Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

  Startpoint: pkg_detect (input port clocked by clk)
  Endpoint: counter/accept_count_reg[15]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  weight_verification_top
                     8000                  saed32rvt_ss0p95v125c

  Point                                                    Incr       Path
  -----------------------------------------------------------------------
  clock clk (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  input external delay                                     0.20       0.20 r
  pkg_detect (in)                                          0.00       0.20 r
  U145/Y (NAND3X0_RVT)                                     0.05       0.25 f
  U104/Y (INVX0_RVT)                                       0.04       0.29 r
  U146/Y (AND3X1_RVT)                                      0.07       0.36 r
  U147/Y (NAND2X0_RVT)                                     0.04       0.41 f
  U148/Y (INVX1_RVT)                                       0.03       0.44 r
  U149/Y (NAND2X0_RVT)                                     0.05       0.48 f
  U150/Y (INVX1_RVT)                                       0.03       0.52 r
  U151/Y (NAND2X0_RVT)                                     0.04       0.56 f
  U152/Y (INVX1_RVT)                                       0.03       0.59 r
  U153/Y (NAND2X0_RVT)                                     0.04       0.64 f
  U154/Y (INVX1_RVT)                                       0.03       0.67 r
  U155/Y (NAND2X0_RVT)                                     0.04       0.71 f
  U156/Y (INVX1_RVT)                                       0.03       0.74 r
  U157/Y (NAND2X0_RVT)                                     0.04       0.79 f
  U158/Y (INVX1_RVT)                                       0.03       0.82 r
  U159/Y (NAND2X0_RVT)                                     0.04       0.86 f
  U160/Y (INVX1_RVT)                                       0.03       0.89 r
  U161/Y (NAND2X0_RVT)                                     0.04       0.94 f
  U162/Y (INVX1_RVT)                                       0.03       0.97 r
  U163/Y (NAND2X0_RVT)                                     0.04       1.01 f
  U164/Y (INVX1_RVT)                                       0.03       1.04 r
  U165/Y (NAND2X0_RVT)                                     0.04       1.09 f
  U166/Y (INVX1_RVT)                                       0.03       1.12 r
  U167/Y (NAND2X0_RVT)                                     0.04       1.16 f
  U168/Y (INVX1_RVT)                                       0.03       1.20 r
  U169/Y (NAND2X0_RVT)                                     0.04       1.24 f
  U108/Y (INVX0_RVT)                                       0.04       1.27 r
  U170/SO (HADDX1_RVT)                                     0.10       1.37 f
  counter/accept_count_reg[15]/D (DFFARX1_RVT)             0.00       1.37 f
  data arrival time                                                   1.37

  clock clk (rise edge)                                    2.00       2.00
  clock network delay (ideal)                              0.00       2.00
  clock uncertainty                                       -0.30       1.70
  counter/accept_count_reg[15]/CLK (DFFARX1_RVT)           0.00       1.70 r
  library setup time                                      -0.05       1.65
  data required time                                                  1.65
  -----------------------------------------------------------------------
  data required time                                                  1.65
  data arrival time                                                  -1.37
  -----------------------------------------------------------------------
  slack (MET)                                                         0.28
```

After optimization

Power Report Analysis

```
Global Operating Voltage = 5
Power-specific unit information :
    Voltage Units = Unitless
    Capacitance Units = Unitless
    Time Units = Unitless
    Dynamic Power Units = Unitless    (derived from V,C,T units)
    Leakage Power Units = Unitless


Attributes
----------
i - Including register clock pin internal power
```

```
  Cell Internal Power  =   0.0000     (0%)
  Net Switching Power  =   0.0265   (100%)
                         ---------
Total Dynamic Power    =   0.0265   (100%)

Cell Leakage Power     =   0.0000

Error: Either dynamic power or leakage power, in the library, is unitless. Unable to display complete power group summary. (PWR-799)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000               NA   (     N/A)
memory             0.0000           0.0000            0.0000               NA   (     N/A)
black_box          0.0000           0.0000            0.0000               NA   (     N/A)
clock_network      0.0000           0.0000            0.0000               NA   (     N/A)  i
register           0.0000           0.0000            0.0000               NA   (     N/A)
sequential         0.0000           0.0000            0.0000               NA   (     N/A)
combinational      0.0000        2.6482e-02           0.0000               NA   (     N/A)
--------------------------------------------------------------------------------------------
Total              0.0000        2.6482e-02           0.0000               NA
1
```

Before optimization

```
----------
i - Including register clock pin internal power


  Cell Internal Power  = 143.2452 uW   (98%)
  Net Switching Power  =   3.4323 uW    (2%)
                         ---------
Total Dynamic Power    = 146.6775 uW  (100%)

Cell Leakage Power     =  34.9458 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network    137.5106           0.0000            0.0000          137.5106  (  75.71%)  i
register           1.5550           0.5583        2.3222e+07           25.3358  (  13.95%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational      4.1796           2.8740        1.1723e+07           18.7770  (  10.34%)
--------------------------------------------------------------------------------------------
Total            143.2452 uW        3.4323 uW     3.4946e+07 pW       181.6233 uW
1
~
```

After optimization

Area Report Analysis

TEAM_EDITH_DESIGN_RPT.rpt

```
****************************************
Report : area
Design : weight_verification_top
Version: W-2024.09-SP5-3
Date   : Sat Mar 14 06:29:53 2026
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    gtech (File: /home/sreenivasulu5137/synopsys/dcs/syn/W-2024.09-SP5-3/libraries/syn/gtech.db)

Number of ports:                          448
Number of nets:                           660
Number of cells:                          243
Number of combinational cells:            158
Number of sequential cells:                58
Number of macros/black boxes:               0
Number of buf/inv:                         23
Number of references:                       5

Combinational area:                  0.000000
Buf/Inv area:                        0.000000
Noncombinational area:               0.000000
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                     0.000000
Total area:                 undefined

Information: This design contains unmapped logic. (RPT-7)
1
```

Before optimization

```
****************************************
Report : area
Design : weight_verification_top
Version: W-2024.09-SP5-3
Date   : Sat Mar 14 06:26:08 2026
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/sreenivasulu5137/POST_OPT_DESIGN/models/saed32rvt_ss0p95v125c.db)

Number of ports:                           78
Number of nets:                           254
Number of cells:                          221
Number of combinational cells:            171
Number of sequential cells:                50
Number of macros/black boxes:               0
Number of buf/inv:                         50
Number of references:                      20

Combinational area:                345.889986
Buf/Inv area:                       66.077440
Noncombinational area:             355.801606
Macro/Black Box area:                0.000000
Net Interconnect area:             157.897592

Total cell area:                   701.691592
Total area:                        859.589184
1
```

After optimization

QoR (Quality of Results) Report Analysis

```
-----------------------------------
Combinational Area:        0.000000
Noncombinational Area:     0.000000
Buf/Inv Area:              0.000000
Total Buffer Area:             0.00
Total Inverter Area:           0.00
Macro/Black Box Area:      0.000000
Net Area:                  0.000000
-----------------------------------
Cell Area:                 0.000000
Design Area:               0.000000


Design Rules
-----------------------------------
Total Number of Nets:           295
Nets With Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0
-----------------------------------


Hostname: sd46.ece.kalasalingam.ac.in

Compile CPU Statistics
----------------------------------------
Resource Sharing:                   0.00
Logic Optimization:                 0.00
Mapping Optimization:               0.00
----------------------------------------
Overall Compile Time:               0.00
Overall Compile Wall Clock Time:    0.00

--------------------------------------------------------------------

Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


Design (Hold)  WNS: 0.30  TNS: 25.80  Number of Violating Paths: 98

--------------------------------------------------------------------
```

Before optimization

```
Macro/Black Box Area:      0.000000
Net Area:                157.897592
-----------------------------------
Cell Area:               701.691592
Design Area:             859.589184


Design Rules
-----------------------------------
Total Number of Nets:           254
Nets With Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0
-----------------------------------


Hostname: sd46.ece.kalasalingam.ac.in

Compile CPU Statistics
----------------------------------------
Resource Sharing:                   0.05
Logic Optimization:                 0.03
Mapping Optimization:               0.16
----------------------------------------
Overall Compile Time:               2.28
Overall Compile Wall Clock Time:    2.56

--------------------------------------------------------------------

Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

--------------------------------------------------------------------
```

After optimization

Constraint & Timing Achievement

- Clock constraint: **2 ns (500 MHz)**
- **Setup timing:** All constraints met (Positive slack)
- **Hold timing:** Minor short paths observed due to counter feedback logic have been solved.
- Zero Violating Paths

```
# Create Clock (500 MHz)
create_clock -name clk -period 2 [get_ports clk]

# Clock Uncertainty
set_clock_uncertainty -setup 0.3 [get_clocks clk]
set_clock_uncertainty -hold  0.1 [get_clocks clk]


# Input Delay
set_input_delay 0.2 -clock clk [get_ports {weight_in[*]}]
set_input_delay 0.2 -clock clk [get_ports {min_weight[*]}]
set_input_delay 0.2 -clock clk [get_ports {max_weight[*]}]
set_input_delay 0.2 -clock clk [get_ports pkg_detect]

# Output Delay
set_output_delay 0.3 -clock clk [get_ports {sort_accept sort_reject}]

# Reset as False Path
set_false_path -from [get_ports rst]

# Output Load
set_load 0.1 [get_ports {sort_accept sort_reject}]
```

Reframed SDC

Conclusion

- Developed an ASIC-based Digital Package Weight Verification System for automated packaging applications.
- The design successfully implements all the expected features required for the system, including:
    - Weight comparison logic using comparator
    - Acceptance or rejection control for decision making
    - Package counters to track total, accepted, and rejected packages
    - Fault alert output to detect abnormal conditions
    - Sorting control signals to direct packages to accept or reject paths
- The system performs real-time weight verification and automatic sorting, improving production efficiency.
- The modular RTL architecture makes the design suitable for ASIC implementation, scalable, and reliable for industrial packaging systems.

Thank You!

K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(Autonomous)

Tiruchengode - 637 215. Namakkal Dt. Tamil Nadu. India